

Концентрация меры на сфере

1 сентября 2026 года

In high dimensions almost all of the measure of a sphere sits near any equator, and a Lipschitz function is almost everywhere equal to its median. Hence the almost Euclidean sections of convex bodies (Dvoretzky) and the whole probabilistic technique of asymptotic geometry.

Notation. σ is the rotation invariant probability measure on S^{n-1} ; for $A \subseteq S^{n-1}$ and $\varepsilon > 0$ let A_ε be the set of points within geodesic distance ε from A .

Theorem 1 (isoperimetry on the sphere). Among all sets of a given measure, the spherical cap minimises $\sigma(A_\varepsilon)$ for every $\varepsilon > 0$.

Theorem 2 (Lévy's lemma). If $\sigma(A) \geq \frac{1}{2}$, then $\sigma(A_\varepsilon) \geq 1 - 2e^{-n\varepsilon^2/2}$. Consequently, for a 1-Lipschitz function $f: S^{n-1} \rightarrow \mathbb{R}$ with median M , $\sigma(|f - M| > \varepsilon) \leq 4e^{-n\varepsilon^2/2}$.

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1. Prove that the cap $\{x \in S^{n-1} : x_1 \geq \varepsilon\}$ has measure at most $e^{-n\varepsilon^2/2}$, and deduce that for large n almost all of the sphere lies within $O(1/\sqrt{n})$ of any fixed equator.
 2. Deduce Theorem 2 from Theorem 1 (which you may use without proof), and give an example of a 1-Lipschitz function on S^{n-1} for which the estimate is sharp up to constants.
 3. (MIPT 2023) Prove that if a set $X \subseteq S^n$ occupies more than half of the Riemannian volume of the unit sphere, then the geodesic segments of length at most π with both endpoints in X cover the whole sphere.
 4. (MIPT 2010) A point x is uniformly distributed on the unit sphere of $\mathbb{R}^n$, and a point y with $\|y\| = 1$ is fixed. Let l_n be the expected distance between x and y . Find $\lim_{n \rightarrow \infty} l_n$.

5. (ФКН, осень 2022) One percent of the surface of a sphere is coloured black; for every black point we colour blue the equator having that point as a pole. Prove that at least ten percent of the surface is coloured blue.
6. (Карасёв 1.87) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be bounded and continuous, and suppose that at every point its value equals its average over the unit sphere centred at that point. Prove that f is constant.